

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
DEPARTMENT OF ELECTRICAL ENGINEERING
MIDTERM EXAMINATION-II: Date: 07-04-2026
B.Tech. 2nd Semester

Subject: Electrical Circuit Theory
Time Allotted: 50 min.

Subject Code: EEPC 101
Max Marks: 15

Answer All the Questions

- Q1. a) If M is the mutual inductance between two coils having self-inductances L_1 and L_2 , what will be the coefficient of coupling? (1M)
- b) Find V_2 in the circuit shown in Fig. Q1b, such that the current in the left-hand loop (loop 1) is zero. Assume: $V_1 = 5\angle 0^\circ$ (4M)
- Q2. a) Define Tree and Co-tree. (1M)
- b) Write the cut-set matrix and tie-set matrix for the network shown, with branches (1, 2, 3) taken as twigs. The branches and nodes are numbered. for the given Network shown in Fig. Q2.b (4M)
- Q3. a) Write the mathematical expressions for the step function and ramp function. (1M)
- b) For the circuit shown in Fig. Q3b, at $t = 0$ the switch is opened. Assume initial conditions: $V_C(0+) = 1\text{ V}$, $i_L(0+) = 1\text{ A}$, $L = 0.5\text{ H}$, $C = 1\text{ F}$, $G = 1\text{ mho}$, $V = 1\text{ V}$. Draw the transformed circuit for $t > 0$ and write the KCL equations at node 1 and node 2. (4M)

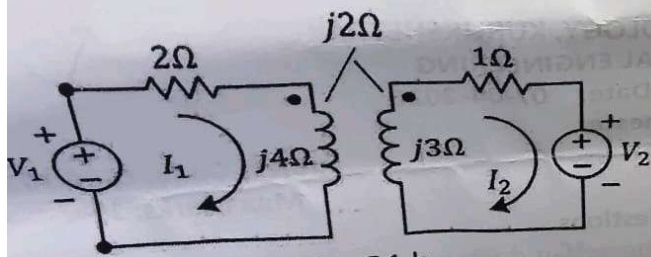


Fig. Q1 b

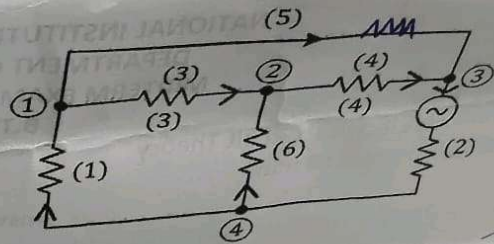


Fig. Q2 b

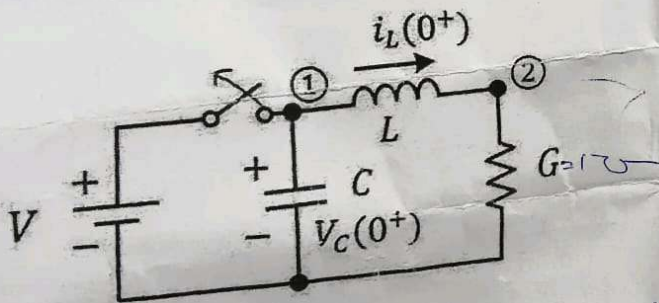


Fig. Q3. b

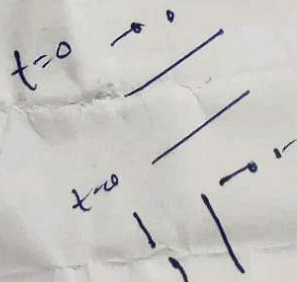
$$-5\angle 0^\circ + 2\angle 71^\circ + 4j\angle 71^\circ - 2\angle 71^\circ = 0$$

$$5 + 2j\angle 71^\circ$$

$$\angle 71^\circ = -\frac{5\angle 0^\circ}{2j}$$

$$V_2 = -2\angle 0^\circ (1 + 3j) = -5$$

$$i_2 + 2j\angle 71^\circ + \angle 71^\circ + V_2 = 0$$



$$V = L \frac{di}{dt}$$

$$V = \frac{d\phi}{dt}$$

$$\frac{dy}{dx}$$

$$\text{der } x^2 + y$$